# Chapter 7: Structures and Unions

In programming, especially in languages like C and C++, structures and unions provide powerful ways to manage and manipulate related data. These constructs allow you to group various types of data together under one name, enhancing code organization, readability, and efficiency. This chapter explores structures, nested structures, unions, and enumerations, explaining their definitions, usage, and key differences.

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## Defining Structures and Accessing Members

### What is a Structure?

A **structure**( struct ) is a user-defined data type that allows you to group variables of different types under one name. Each element in a structure is called a **member**. Structures help manage related information more efficiently, especially when dealing with complex data types.

### Defining a Structure

Here's how to define a structure in C or C++:

#include <stdio.h>